

SSI INDEX

Who, What, Where, When, Why, How

Building the Foundation for Open Infrastructure Intelligence
across 38 OECD Nations

Ikenga | Employee-Owned | Madrid & Naples

April 2026

Why

The problem no one is measuring — and why it matters now

The Same Outage. Two Realities.

Every version of the SSI Index reveals a deeper layer of the same injustice

v4.0.2 POWER

Current

MILAN

A 48-hour blackout in Milan is a logistical challenge. Backup generators kick in. Businesses switch to remote. Life goes on.

CALABRIA

The same blackout in Calabria means hypothermia for elderly residents, spoiled medication, closed schools, and businesses that never reopen. Energy poverty: 17.8%.

Yet every grid framework rates these substations as equal risks.

v4.5 POWER + WATER

End 2027

MILAN

When Milan loses power, redundant water systems and backup pumps maintain pressure. Hospitals keep sterile water. The vaccine cold chain holds.

CALABRIA

When Calabria loses power, it silences the pumps. Water pressure drops within hours. The dialysis centre loses sterile water. The cold chain breaks for thousands of doses.

A power failure becomes a public health emergency — and no framework models that coupling.

v5.0 POWER + WATER + COMPUTE

End 2028

MILAN

Milan has three data centres, redundant fibre, and edge compute within 5 km. Telemedicine, dispatch, and digital government continue uninterrupted.

CALABRIA

Calabria loses connectivity through the failed substation. Telemedicine to the hospital 40 min away — gone. Emergency dispatch — gone. Digital services that replaced closed offices — gone.

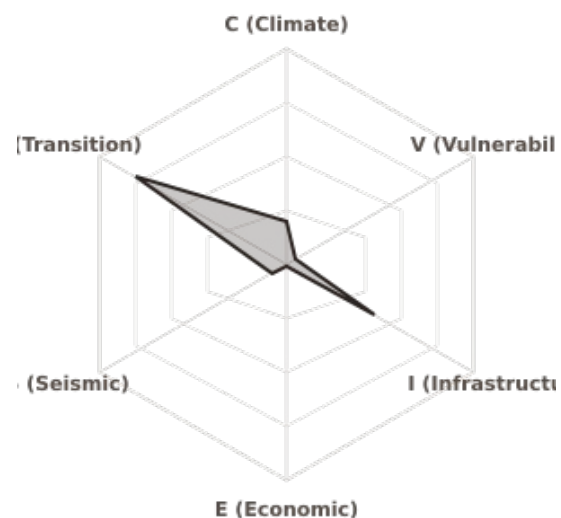
That's not a grid failure. That's a community going dark in every sense of the word.

The Data Behind the Narrative

Live SSI v4.0.2 data — 57 substations in Milano vs 106 in Calabria

MILANO

57 substations | Lombardia | $\bar{R} = 0.31$

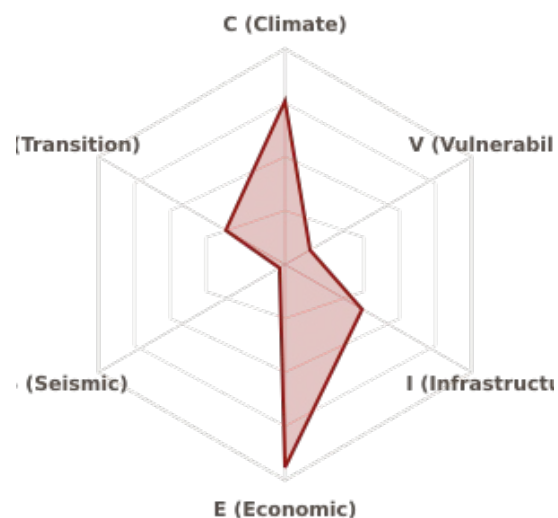


CSRD E1-E5 | SDG 7, 9, 11 | SFDR PAI

€52,700	GDP per capita	5.2%	Unemployment
5.5%	Energy poverty	+5.2	Migration score
1.36%	R&D / GDP	0%	High-risk substations

CALABRIA

106 substations | 5 provinces | $\bar{R} = 0.59$



CSRD E1-E5 | SDG 1, 7, 10, 11 | SFDR PAI

€16,245	GDP per capita	20.0%	Unemployment
17.0%	Energy poverty	-4.8	Migration score
0.52%	R&D / GDP	93.4%	High-risk substations

Source: ikengassiindex.github.io/italy | SSI v4.0.2 | 95 variables | 10,000 Monte Carlo iterations | CC BY-SA 4.0

Why This Question Is Fundamental

Six structural reasons why invisible infrastructure inequality must be addressed now

01 Energy Sovereignty Requires Granular Visibility

National grid resilience demands understanding of every critical node. When substation-level vulnerability is invisible, nations cannot assess whether energy independence is genuine or concentrated in geographically exposed corridors.

OECD, Electricity Grids & Secure Energy Transitions (2023); IEA: only 18% of countries have concrete electricity climate-resilience plans (2024)

02 Rural Depopulation Cascades into Wildfire Risk

Abandoned farmland eliminates natural firebreaks — vineyards, grazed pastures, managed forests. Depopulation hollows the human stewardship that suppresses ignition, while ageing grid assets in emptying regions receive less maintenance.

EU Integrated Wildfire Risk Management (2024); EU wildfires burn 500k+ hectares/year

03 Just Transition Obligations Demand Equity Mapping

The green transition risks displacing workers and harming local economies if substation-level exposure remains unmapped. Without granular data, just-transition funds flow to political lobbies rather than to the communities most at risk.

IPCC AR6 WG3, Ch. 17: Accelerating the Transition in Context of Sustainable Development (2023)

04 Territorial Cohesion Drives Economic Competitiveness

Metropolitan regions command ~32% higher GDP per capita than rural hinterlands. Invisible infrastructure inequality widens this gap — deterring investment, accelerating brain drain, and eroding the fiscal base of subnational governments.

OECD Regional Outlook (2023); OECD Regions & Cities at a Glance (2024)

05 Climate Adaptation Has a 1:6 Cost Multiplier

Every euro invested in climate-resilient infrastructure saves six euros in avoided losses. Delayed action nearly doubles costs over a decade — but you cannot adapt what you cannot see at the asset level.

OECD Infrastructure for a Climate-Resilient Future (2024); IEA Climate Resilience Report (2024)

06 The Digital Divide Rides on the Electrical Grid

Rural broadband piggybacks on electricity networks — just as it did in 1930s rural electrification. OECD data shows rural residents face 44% slower broadband and only 78% access to 30 Mbps. Unmapped grid inequality blocks coordinated infrastructure deployment.

OECD Digital Connectivity (2025); OECD Closing Broadband Divides (2024)

The Opportunity

Infrastructure investment decisions are blind to human vulnerability

Across the OECD, 205,000+ electrical substations

serve as the invisible backbone of modern life. Yet every major grid risk framework rates a substation in prosperous Milan the same as one in energy-poor Calabria — where 17.8% of households can't afford heating.

The SSI Index is the first open digital twin fusing core infrastructure risk with societal exposure at substation resolution.

We call this ***Breaking the Silos*** — the moment when grid engineering, social policy, and climate adaptation converge into a single decision framework.

Our Italy baseline proves it works: 4,293 substations, 95 variables, 42.9M Monte Carlo simulations, and 89% alignment with utilities' own post-hoc prioritisation.

205k+

OECD Substations

1.4B

People Served

91/100

Competitive Score

€13

Cost per Substation

What

The SSI Index — six components, one auditable score

Why We Exist

PURPOSE

To make infrastructure vulnerability visible at the substation level — so that decisions about energy, water, and digital access are informed by where people actually live, not where capital already flows.

MISSION

Produce and maintain an open, auditable index of infrastructure resilience across the OECD — covering power, water, and compute systems — that enables regulators, utilities, investors, and communities to identify, measure, and act on socioeconomic disparities in critical service delivery.

VISION

An OECD where every substation-connected community has equal visibility in the frameworks that allocate capital, plan resilience, and govern the green and digital transitions — from Milan to Calabria, from Tokyo to Tromsø.

THEORY OF CHANGE

By coupling Monte Carlo risk modelling with ESG-grade reporting at substation resolution, the SSI Index converts invisible infrastructure inequality into auditable data — giving regulators the evidence to act, funders the metrics to invest, and citizens the transparency to hold systems accountable.

Six Components, One Composite Score

Why these dimensions — and why no competing framework covers all six

C	Climate Wildfire, flood, heat-wave, wind, ice-storm probability at the asset site <i>Climate hazard is the dominant long-term risk driver — unpriced by traditional reliability metrics</i>
V	Vulnerability Elderly share, energy poverty, migration loss, demographic fragility <i>A substation serving a fragile community fails differently — human exposure must weight the score</i>
I	Infrastructure Asset age, capacity headroom, redundancy (N-1), maintenance backlog <i>Engineering condition is necessary but not sufficient — it is the only dimension legacy indexes cover</i>
E	Economic GDP per capita, unemployment, R&D intensity, fiscal capacity of the served area <i>Economic context determines recovery speed — wealthy areas self-insure, deprived areas do not</i>
S	Seismic Peak ground acceleration, fault proximity, soil amplification, liquefaction susceptibility <i>Seismic risk is catastrophic and correlated — a single event can cascade across an entire grid zone</i>
T	Transition Renewable penetration, EV load growth, grid modernisation, policy readiness <i>The green transition is reshaping load profiles — substations that cannot adapt become bottlenecks</i>

COVERAGE MATRIX				
SSI Index	SAIDI / SAIFI	CDP	MSCI ESG	GRESB
●	—	●	●	●
●	—	—	—	—
●	●	—	—	●
●	—	—	●	—
●	—	—	—	—
●	—	●	●	●

● Full coverage ● Partial — Not covered

The SSI is the only framework that fuses all six dimensions into a single auditable score — competitors cover at most two.

Six ESG Reports from One Engine

Native ESG intelligence — not a compliance overlay, but integrated into every risk score

R1

SDG 13

SDG 9

SDG 11

Climate Physical Risk

ESRS E1 · TCFD · Taxonomy Annex A

CMIP6 projections, seismic hazard, corrosion indices at 250m resolution under SSP2–4.5 and SSP5–8.5 trajectories

R2

SDG 7

SDG 10

SDG 1

Grid Equity & Social Impact

ESRS S1/S2 · SFDR PAI · UN PRI

Energy poverty, demographic fragility, critical facility proximity — the “Breaking the Silos” metric

R3

SDG 9

SDG 13

SDG 12

EU Taxonomy Alignment

EU Taxonomy Art. 11 · DNSH · Min. Safeguards

Substantial contribution, DNSH, minimum safeguards at individual substation resolution

R4

SDG 7

SDG 13

SDG 9

Energy Transition Readiness

ESRS E1 Transition · TCFD · Green Deal

DER penetration, storage capacity, demand-response, EV density, smart meter deployment

R5

SDG 11

SDG 3

SDG 15

Pollution & Corrosion

ESRS E2 · ISO 9223 · CIGRE TB 761

ISO 9223, CIGRE TB 761 Markov degradation, volcanic/industrial exposure

R6

SDG 9

SDG 16

Cybersecurity Exposure

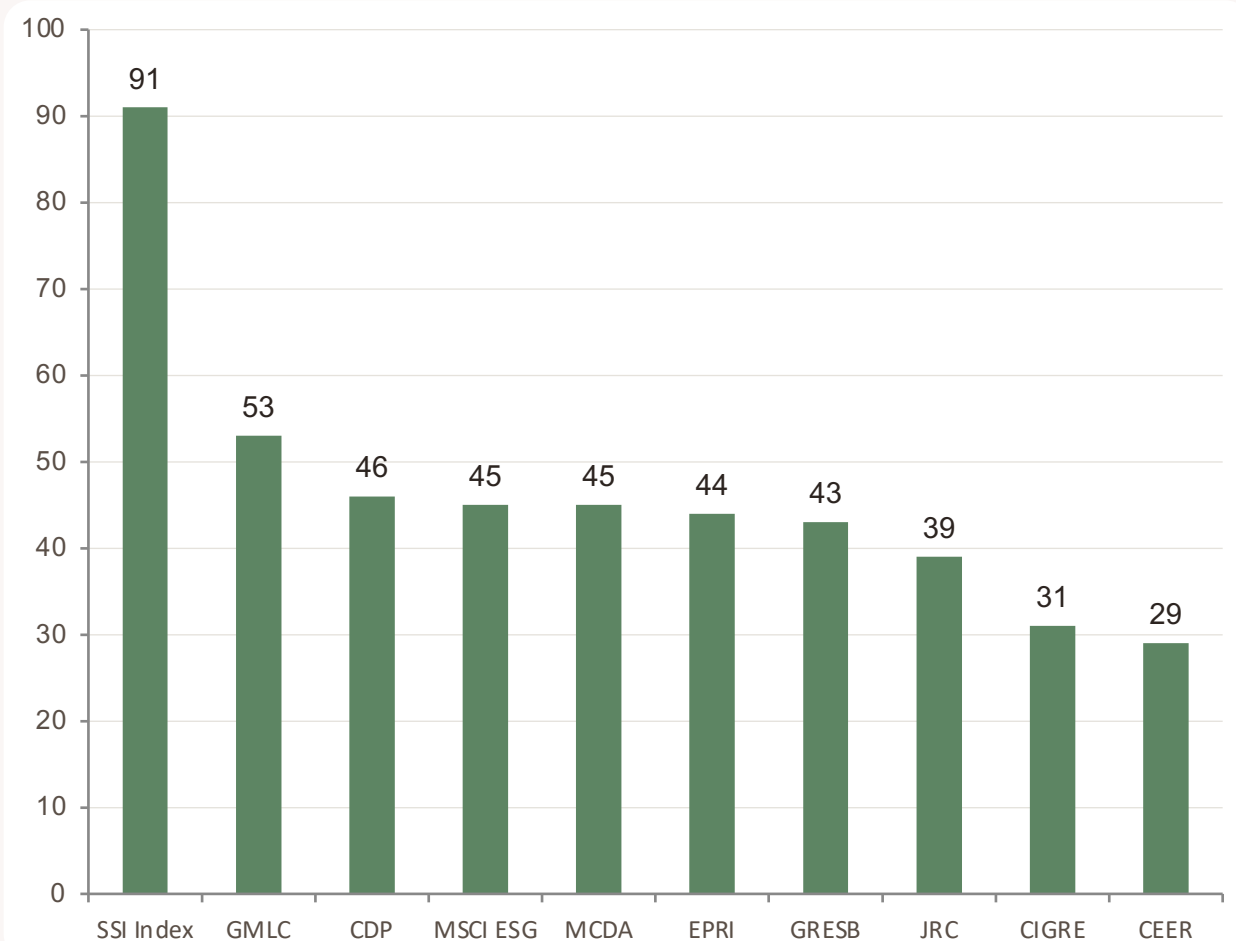
NIS2 · ENISA · ESRS G1

SCADA connectivity, NIS2 classification, digital twin attack surface, network topology

Each ESG report is designed to CSRD Article 29a limited-assurance standard

Competitive Benchmark

91/100 across 20 dimensions — leading on 19 of 20 against all engineering and ESG peers



Key Differentiators

- Only framework fusing engineering risk + ESG at substation level
- Full Monte Carlo at fleet scale (10k iterations per asset)
- Open data: 95 vars, 30+ sources, zero proprietary dependency
- 6 CSRD-auditable ESG reports per substation from same engine
- Cost: €13/substation vs. €840–€10,459 for peers

The Seventh Axis.

Because at the end of every substation there is a community.

V (Vulnerability) already captures energy-poverty rate, elderly share and demographic fragility at the region level. H (Human) extends the SSI with ten community metrics measured at the substation catchment — turning a risk score into a direct statement about who it serves and what a failure costs them.

LIFE & CARE

H1 Life-Support Load

People in the catchment reliant on mains-powered medical devices (dialysis, oxygen, CPAP, insulin pumps).

H2 Cold-Chain Clinical Nodes

Count and type of time-critical facilities served: ICU/NICU, dialysis centres, blood banks, vaccine depots.

H3 Child-and-Elderly Thermal-Risk Pop.

Absolute count of residents under 5 and over 75 — the cohorts most sensitive to heat and cold stress.

LIVELIHOOD & AUTONOMY

H4 Energy-Poverty Severity

Depth (not rate): % of income spent on energy by the bottom quartile, plus arrears-on-bills ratio.

H5 Home-Work Inflexibility

Share of workforce in occupations that cannot be done remotely (ISCO 6–9) — converts outage hours into lost wages.

H6 Caregiving Burden Index

Single-adult households with care dependents (<5, >75, or disabled) — surfaces the gendered cost of grid failure.

VOICE & CONNECTION

H7 Digital-State Exposure

Share of essential public services (tax, health, welfare, school) that are digital-only when the grid fails.

H8 Linguistic & Info-Access Gap

Adults with limited host-language proficiency or low digital literacy — miss emergency alerts and updates.

H9 Civic Infrastructure Density

Third-places per 10k residents (libraries, community centres, parish halls, pharmacies) — ad-hoc relief hubs.

H10 Food-System Fragility · Refrigerated food-retail and food-service employment per 1,000 residents — a proxy for hyperlocal economic loss per outage hour.

Where

From 23 + 1 today to 38 OECD nations

Where We Are Today

23 countries + 1 territory operational — the largest open grid intelligence platform in the world

23+1

Countries +
Territory

7

Academic Papers
Submitted

6

ESG Reports
per Substation

70

Copyright
Registrations*

** 3 registrations per country: Technical Reference Document, Competitive Landscape Report, Source Code Deposit*

Dashboard Live

23 countries + 1 territory on GitHub Pages with full ESG analytics, radar charts, and intelligence summaries per country.

Proven Methodology

95 variables, 30+ data sources per country, 10,000 Monte Carlo iterations per substation. 89% alignment with utility prioritisation lists.

Foundation Ready

Non-profit governance designed under Italian law. Leadership designated: Davide de Caro (Chairman), Vjekoslav Bonic (Advisory Board Head).

Who

The team, the credibility, the network

Academic & IP Credibility

Peer-reviewed rigour and comprehensive intellectual property protection

7 Peer-Reviewed Papers Submitted

- Energy Policy (IF ~9.0)
- Energy Research & Social Science (IF ~8.5)
- Climate Policy (IF ~7.1)
- Sustainable Energy, Grids & Networks (IF ~5.4)
- J. Modern Power Systems & Clean Energy (IF ~5.5)
- Utilities Policy (IF ~5.4)
- J. Infrastructure Intelligence & Resilience (Elsevier)

First acceptances expected June–September 2026

Intellectual Property

- 70 USCO & SIAE copyright registrations (3 per country × 23 + 1 ESG formula construct):
 - Technical Reference Document • Competitive Landscape Report • Source Code Deposit
- Scales to 117 copyrights at 38-OECD expansion
- Berne Convention protection across 181 signatory nations
- Patent pending: Kinetic Shield™ (ES, JP, US, UK)
- SSI-ENN© AI grid intelligence platform
- All outputs under Creative Commons BY-SA 4.0

When

The timeline and what success looks like

Growth Trajectory

From 23 + 1 today to 38 OECD nations by end 2026

TODAY

Q2 2026

- 23 + 1 territory live
- ~150k+ substations
- 7 papers submitted
- Dashboard operational
- 6 ESG reports per asset

H2 2026

Q3–Q4

- Remaining 14 OECD countries
- ~180k+ substations
- First acceptances expected
- Advisory board convenes
- External audit of baseline

2027

Year 2

- 38 OECD nations complete
- 205k+ substations
- University partnerships (6+)
- Licensing pilots launch
- 500+ workshop participants

2027+

Year 2+

- v4.5: Power–Water coupling
- v5.0: Power–Water–Compute
- Self-sustaining revenue
- 1.4B people covered
- Board expansion to 12

What Success Looks Like

Key performance indicators — end of 2027



38

OECD Countries
Fully Operational



205k+

Substations
Assessed



€500k+

Foundation Grants
Secured



5+

Utility MOUs
Signed



6+

University
Partnerships



10+

Published
Academic Papers



3+

Utilities Using SSI
in CSRD Reporting



500+

Workshop
Participants / Year

Deliverables Framework

What the SSI Foundation delivers, to whom, and how often — aligned with market practice

RESEARCH & POLICY COMMUNITY		
SSI Flagship Report	Annual (Q4)	Full 23 + 1 territory infrastructure resilience assessment
Peer-Reviewed Papers	2-4 / year	Submitted to IEEE, Nature Energy, Applied Energy, etc.
Policy Briefs	2-3 / year	Targeted at ACER, ENTSO-E, national regulators, EU DG ENER
Open Dataset Release	Annual	Zenodo/GitHub deposit with DOI — CC BY-SA 4.0 licence
INSTITUTIONAL PARTNERS		
Advisory Board Meetings	Quarterly	Strategic alignment, partnership pipeline review
Annual Stakeholder Workshop	Annual (Q2)	In-person event with utilities, regulators, academia
Conference Presentations	3-5 / year	IEEE PES, CIGRE, CIRED, EU Sustainable Energy Week
Country Spotlight Reports	2-4 / year	Deep-dive national reports for partner engagement
PUBLIC & MEDIA		
Press Releases & Op-Eds	4-6 / year	Timed to flagship publications and policy windows
Newsletter	Monthly	Dashboard updates, research highlights, event calendar
Dashboard Updates	Continuous	Live data refreshes on public SSI dashboard

Cadence modelled on Rockefeller Foundation grantee reporting, IRENA/IEA annual cycles, and GRESB benchmark publication schedules.

Today, Calabria loses power and the lights go out.

Tomorrow, it loses water and the pumps fall silent.

By 2028, it loses compute and the community goes dark
in every sense of the word.

Milan will always be fine. The SSI Index exists for Calabria — and for every community across the OECD where vulnerability is invisible to the frameworks that allocate capital.

The SSI Index exists to make 1.4 billion people visible to the decisions that determine whether their lights stay on, their water flows, and their communities remain connected.

Infrastructure that gives back. Employee-owned. Sovereign by design.

Ikenga

Infrastructure that gives back.

ssi_index@ikenga.eu | ikenga.eu | ikengassiindex.github.io